WORKSHOP PRINT PACK · PROTOTYPE DRAFT

DE PAPIEREN MACHINE

THE PAPER MACHINE

A simple next-word model, built by hand out of the KB's own newsprint — no screens, no power, no software — leaving two public paper archives behind: what was excluded, and what stayed available but unsaid.

Duration	2 hours 30 (150 min), one break
Participants	Designed for 8–10 · maximum 12 · two tables
Facilitators	One. No assistant required.
Language	English facilitation · Dutch material, glossed in English
Space	KB Atelier, ground floor, window-facing
Leaves behind	TWEE SOORTEN STILTE — two window lines, growing across the run

WHAT IS IN THIS PACK

p.2 — Operator's manual · three operations, one parallel operation
p.3 — Station 1 · CORPUS
p.4 — Station 2 · TOKENISE
p.5 — Station 3 · CLASSIFY (parallel)
p.6 — Station 4 · PREDICT
p.7 — Two kinds of silence · take-home · sources
p.8 — Reading-window masks · facilitator pre-cuts the slots
+ separate file — **transcription sheet**, A4 landscape, printed on 2-part NCR
p.9 — Collection-policy cards
p.10–13 — Vocabulary sheets A / B / C / D
p.14–15 — Carrier cards · EXCLUDED and AVAILABLE
p.16–17 — Window headers · print at A3
p.18–21 — Station signs 1–4
p.22 — Excluded box label
p.23 — Take-home colophon cards
p.24 — Facilitator cue sheet · the only page you hold during the session

YOU ARE THE MACHINE TODAY

Today you build a deliberately simple next-word model — a **bigram model** — and expose the human decisions around it. Not a small chatbot: no memory beyond one word, no meaning, no training objective. The smallest thing that can honestly be called a statistical language model, and small enough that eight people can run it with scissors.

THREE OPERATIONS BUILD THE MODEL

1 · CORPUS → what goes in, and what never does
2 · TOKENISE → text becomes countable pieces
4 · PREDICT → sample the next word by frequency

ONE PARALLEL OPERATION, ON THE SAME MATERIAL

3 · CLASSIFY → what can be found, filtered, governed
— it does not build the model

THE STATIONS RUN 1 → 2 → 3 → 4 IN THE ROOM. STATION 3 SITS APART HERE BECAUSE IT FEEDS A DIFFERENT SYSTEM.

Classification is a real operation — most of what a library does, and much of what dataset owners do. It just builds something else: an index, a filter, a set of permissions. Keeping it visibly separate is the point, because public conversations about AI often run the two together.

THE MATERIAL

Printed facsimiles of Dutch newspaper excerpts, 1618–1879, digitised by the KB and published on Delpher. Publications older than 140 years are public domain: you may cut them, mistranscribe them and reassemble them wrongly. Each carries its date and Delpher identifier.

IF YOU DON'T READ DUTCH

You don't need it. Today's operations are scanning, segmenting, counting and sampling. The classification vocabularies are glossed in English. Sit at either table.

THREE RULES

1. **Nothing goes in the bin.** Rejected material goes in the EXCLUDED box; unused material stays in the tail box. The last half hour is about that difference.
2. **Do not fix your mistakes.** The model is built from your handwriting, errors and all.
3. **Write your rule down before you apply it.** A rule you can write down is a rule somebody else can argue with.

THE DECISION YOU ALREADY MADE

At the door you were handed an excerpt and asked to keep or exclude it, with no context and no rule. That was a **preliminary selection, not part of the model** — neither tray feeds the corpus, and both tables start from the same twenty excerpts whatever you chose.

It returns at the end as evidence of something else: how differently a room decides *before* and *after* a policy has been written down.

WHAT YOU TAKE HOME

One word-pair drawn from the finished machine, and a colophon card. Your transcription becomes the model and is cut up; your two carrier cards stay pegged in the window until the end of January.

WHAT THIS DOES NOT DEMONSTRATE

No attention, no embeddings, no learned weights, no training loop. It cannot show how a modern system represents meaning, and proves nothing about one. What it does show at 1:1 scale are the decisions surrounding *any* model: what was collected, how it was segmented, and who described it.

STATION 2

TOKENISE

30 MINUTES · INDIVIDUAL, THEN TABLE · OPERATION 2 OF 3

Everything the model knows comes from this station. Not from the printed page — from your handwriting. Whatever you get wrong here, the machine will say later.

PART A — BE THE OCR (16 MIN)

Take your excerpt, a **reading window** with the slot already cut, and a **transcription sheet** — forty numbered cells on two-part carbonless paper.

- Hold the window against the column. Read only what appears in the slot, moving **left to right, top to bottom**. No going back. No looking at the whole excerpt.
- **One word per cell.** You decide what counts as a word — a word broken across a line in the original, an abbreviation, a price, a date. That decision is the tokenisation. Nobody will correct it.
- Press firmly: the carbon copy underneath is cut differently and you need both.
- You will not finish. Nobody does. The model uses whatever exists when the timer stops.

AFTER THE TIMER — CHECK ONE ROW AGAINST YOUR PRINTED EXCERPT:

CELLS FILLED	WORDS WRONG	WORDS YOU HAD TO JUDGE

Delpher's search runs on machine OCR of the same material and many errors remain in it — nineteenth-century type, broken columns and damaged paper are hard for software too. Keep your printed excerpt: it is the only clean copy in the room.

PART B — CUT IT INTO PAIRS (10 MIN)

Cut on the printed guides — no measuring, no freehand.

- **Top copy:** cut at the orange A marks → pairs 1+2, 3+4, 5+6 …
- **Carbon copy:** cut at the teal B marks → pairs 2+3, 4+5, 6+7 …
- Cut along every row line too. The **ten single-cell offcuts** → EXCLUDED box.

A full sheet gives **35 usable bigrams**: four pairs from the top copy and three from the carbon, per row, five rows. Each scrap is one bigram. Everything into the tray.

What the cutting rule could not use — two different things.

Four transitions are missing — 8→9, 16→17, 24→25, 32→33. Those words sat next to each other in your reading, but they are on different rows and no scissors will make them one scrap. They are genuinely absent from the model.
Ten single-cell offcuts come off the carbon copy at cells 1 and 8 of each row. Those words *do* appear in pairs elsewhere; the offcuts are redundant. Both go in the excluded box, and they record what the rule could not use — the same artefact that breaks text across columns and pages in every digitised archive.

PART C — LOOK AT WHAT YOU MADE (4 MIN)

QUESTION	ANSWER
Roughly how many pairs does your table hold?	
Which word did you have to make a judgement call about?	
Which of your own errors would you most like to take back?	

STATION 3

CLASSIFY

14 MINUTES · WHOLE ROOM, FOUR GROUPS · THE PARALLEL OPERATION

We are pausing the model to operate a different information system on the same material. Nothing here changes a word the paper machine will say. That is not a defect — it is what this fourteen minutes is for. Classification decides what can be *found*, filtered and governed; it does not decide what comes next.

Leave your tables. The room splits into **four groups of two or three**. Everyone works on the **same single document**, and each group gets one system and may use only its terms.

GROUP A	KB-Code era — the KB's own keyword system, in use **1921–1992**
GROUP B	Brinkman era — Depot keywords, in use **1974 → today**
GROUP C	GOO + Basisclassificatie — in use **1980s–2012**
GROUP D	A 2026 annotation queue — 20 s, paid per item

A–C refer to documented KB systems. Sheets A and B are provisional workshop reductions, to be revised with KB expertise. D is a fictional composite.

ROUND 1 — DESCRIBE IT, IN YOUR GROUP (6 MIN)

Agree one description using only your sheet. If nothing fits, you still have to choose — that is the job.

Group D works differently: one person annotates against the twenty-second sand timer while the rest **watch**. The observers' task is the real one — note what gets skipped, guessed, or forced into whichever field will accept it.

OUR GROUP	OUR DESCRIPTION	WHAT WOULDN'T FIT
A / B / C / D		

In one line: **what would this description make findable, and what would it hide?**

ROUND 2 — READ TO THE ROOM, IN ORDER (6 MIN)

1921, then 1974, then the 1980s, then 2026. Four descriptions of one document. None is wrong. All four then pin to the **OTHER DESCRIPTIONS** card — never the EXCLUDED box, because they were never in the model to begin with.

Three schemes accumulated in one catalogue. KB-Code, Brinkman and GOO overlapped in time and did different jobs; the catalogue still carries all three, because old descriptions stop being applied rather than being rewritten. What you can find depends on which scheme was running when a thing arrived.

Now look back at the sorting board. The contrast becomes visible: not one of these four descriptions is on it. Both systems are real, both matter, and they are not the same system.

STATION 4

PREDICT

30 MINUTES · TABLE, THEN ROOM · OPERATION 3 OF 3

PART A — BUILD IT (8 MIN)

File every scrap by its **first** word. Twelve common words have cells on the board — that is the **head**, and it grows thick. Every other first word goes into the **tail box**, in a rough alphabet bin. Stack loose so thickness shows frequency. Don't tidy.

This is a table of transition counts. **Markov built the first one by hand in 1913** — 20,000 letters of *Eugene Onegin*, counted as vowel–consonant pairs, to prove the letters are not independent. Same operation, with words.

PART B — RUN IT AT YOUR TABLE (8 MIN)

1. **The facilitator announces one start word. Both tables use it.** Write it at the left of the output strip — the two runs have to be comparable.
2. Find that word — **board first, then the tail box**. If it is a tail word: go to its alphabet bin, gather every scrap beginning with that exact word, and draw blindly from those matches only — never from the whole bin.
3. Dot the corner of that scrap immediately — you will not remember later which of two identical scraps you used — then write the second word, replace the scrap, look up the new word, repeat.
4. Stop at 15 words or when a word is in neither. Record where it died, then start again from a different word. **Expect dead ends** — your model is very small.

OUR BEST RUN:

_____

RUNS ATTEMPTED: ____ · LONGEST: ____ WORDS · DIED AT: _____

PART C — THE POOLING EXPERIMENT (12 MIN)

First, predict. Circle what merging both tables will do to the longest run:

LONGER · SHORTER · NO RELIABLE DIFFERENCE → because _____

Then merge, physically. Table B carries its head stacks over and drops each into the matching cell on Table A's board. **Table A** merges the six corresponding tail bins. Two minutes, nobody in anybody's way. Sheets are marked **A** or **B**, so provenance stays visible in the merged stacks.

Then test it: start from the **same word the facilitator gave you at the beginning**, and run the pooled model **three times**.

POOLED LENGTHS: ____ / ____ / ____ · TABLE RUNS WERE: ____ / ____

_____

More data gives more possible transitions per word, but guarantees nothing about any single run. **Whether your prediction held is the finding, either way.**

PART D — READ IT OUT (2 MIN)

Read the longest pooled run aloud; circle a word from a transcription error. And be careful with "lost" from here on: every **undotted** scrap, board or box, stayed available the whole time. It simply wasn't drawn — not the same as the excluded box, which never had a chance at all.

A demonstration from 1948. Building on Markov, Shannon did this by hand — open a book at random, find a word, hunt for it elsewhere, write down what follows: "THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHARACTER OF THIS POINT IS THEREFORE ANOTHER METHOD…"

TWO KINDS OF SILENCE

20 MINUTES · EVERYONE · THEN 8 MINUTES TOGETHER

Your machine stayed silent about nearly everything, for **two different reasons**. They go on two different lines.

LINE 1 · EXCLUDED

Never entered the model. It had no chance.

- every excerpt rejected at Station 1
- 4 missing transitions + 10 offcuts
- the door tray: EXCLUDED WITHOUT A RULE

LINE 2 · AVAILABLE

Entered the model. Never drawn. Present the whole time.

- every undotted scrap, board or box
- everything that stayed available, undrawn

Exclusion is a decision somebody made, and you can name who and when. **Non-selection is a probability** — the material is present, simply not chosen. Systems get attacked for the first and defended with the second, and the two are swapped more often than is honest.

And the other door tray: KEPT WITHOUT A RULE. Hold those up against your table's written policy. Would the rule you wrote at 15:20 have kept them? That gap — between what a room chooses on instinct and what it chooses once a policy exists — is the whole reason we made you write the rule down first.

MOUNT ONE OF EACH

Take one object from the excluded pile, and one **undotted** scrap from anywhere in the model — a head stack on the board or a tail bin, both count. Attach each to its **carrier card** — pink for excluded, teal for available — and answer the question printed on it. Write large: these are read from the street.

PINK — WHICH DECISION REMOVED THIS?

_____

TEAL — WHAT MADE THIS POSSIBLE, BUT UNSELECTED?

_____

Peg both to their lines. They stay in the window until the end of January, beside the cards of everyone who came before you. **Take home instead:** one word-pair drawn from the pooled machine, and the colophon card.

THE LAST QUESTION

You have now performed four operations by hand. Which would you automate, which would you assist, and which would you keep human — and who should decide?

Newspapers 1618–1879 · Delpher / KB, public domain — kb.nl/en/research-find/datasets/delpher-newspapers
KB founded 1798; one copy of every Dutch publication since 1974 — kb.nl/en/over-ons/geschiedenis
KB keyword systems (KB-Code, Brinkman, GOO) — kb.nl/onderzoeken-vinden/bronnen/kb-catalogus/trefwoorden
Nederlandse Basisclassificatie, 4th rev. ed. — kb.nl/kbhtml/docs/bc04.pdf
A. A. Markov, statistical investigation of the text of Eugene Onegin, presented 23 January 1913
C. E. Shannon, A Mathematical Theory of Communication, 1948
P. Otlet & H. La Fontaine, Répertoire Bibliographique Universel, 1895–1934

Vocabulary sheets A and B are provisional workshop reductions, not the thesauri. Sheet D is fiction.

STATION 2 · TOKENISE · PRINT ON 2-PART CARBONLESS (NCR) A4, OR PLAIN PAPER OVER CARBON PAPER

TABLE **A** / **B** (circle one — provenance)

TRANSCRIPTION SHEET · DE PAPIEREN MACHINE

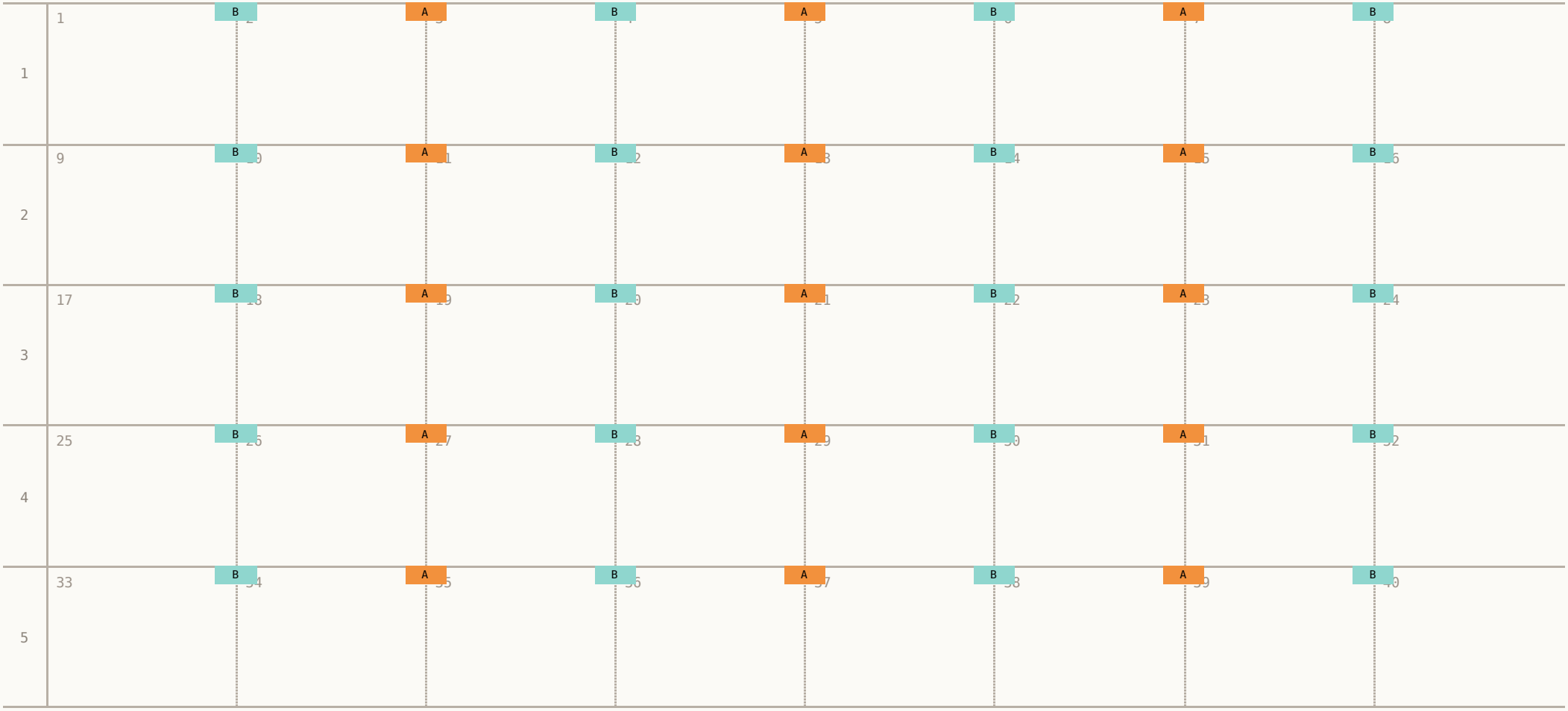
EXCERPT No. _____ NAME _____

ONE WORD PER CELL. You decide what counts as a word. A word broken across a line in the original, an abbreviation, a price, a date — you choose, and your choice is the tokenisation. Nobody will correct it.

LEFT TO RIGHT, TOP TO BOTTOM. Read only through the slot in your reading window. Never move it back. Do not fix mistakes: this handwriting is the entire training corpus.

YOU WILL NOT FINISH. Nobody does, and that is designed in. The model uses whatever exists when the timer stops. Leave the rest empty.

THE CARBON COPY MATTERS. Press firmly. The sheet underneath is cut differently from this one, and the two together give **every adjacent pair within each row**.



CUTTING (10 MIN):

TOP COPY → cut only at A marks → pairs 1+2, 3+4, 5+6 …

CARBON COPY → cut only at B marks → pairs 2+3, 4+5, 6+7 …

Also cut along every row line.

Single-cell offcuts (10 per sheet) → EXCLUDED box.

WHAT THE CUTTING RULE CANNOT USE. The protocol omits **four cross-row transitions** — 8→9, 16→17, 24→25, 32→33 — and produces **ten single-cell offcuts**, at cells 1 and 8 of each row on the carbon copy. These are two different things. The offcuts are redundant copies of words that do appear in pairs elsewhere; the four transitions are genuinely absent from the model. Both go in the excluded box: they record what the cutting rule could not use. The same artefact breaks text across columns and pages in every digitised archive.

A FULL SHEET → **35 USABLE BIGRAMS** · 4 MISSING TRANSITIONS · 10 OFFCUTS

SORTING BOARD · TABLE _____ · OPERATION 3 OF 3

DE PAPIEREN MACHINE

FILE EVERY PAPER PAIR IN THE CELL MATCHING ITS FIRST WORD.
STACK LOOSE — THICKNESS = FREQUENCY. THIS IS THE MODEL.
WHEN POOLING, EACH PERSON MERGES TWO OR THREE HEAD STACKS.

de FOLLOWED BY…	het FOLLOWED BY…	een FOLLOWED BY…	van FOLLOWED BY…
in FOLLOWED BY…	en FOLLOWED BY…	te FOLLOWED BY…	dat FOLLOWED BY…
is FOLLOWED BY…	op FOLLOWED BY…	voor FOLLOWED BY…	den FOLLOWED BY…

THE TAIL → tail box

A–D · E–H · I–L · M–P · Q–T · U–Z

EVERY OTHER FIRST WORD GOES HERE, ONE SCRAP AT A TIME.
BROKEN HALF-WORDS AND UNREADABLE SCRAPS → EXCLUDED BOX.
WHEN YOU GENERATE: BOARD FIRST, THEN THE BOX.

PRINT AT A2 IF POSSIBLE · A3 WORKS FOR A TABLE OF FOUR TO SIX

HEAD: 12 WORDS ON THE BOARD · TAIL: EVERYTHING ELSE, IN A BOX · DOT EVERY SCRAP AS YOU DRAW IT